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Modeling Time-Varying Random Objects and Dynamic Networks

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ABSTRACT

Samples of dynamic or time-varying networks and other random object data such as time-varying probability distributions are increasingly encountered in modern data analysis. Common methods for time-varying data such as functional data analysis are infeasible when observations are time courses of networks or other complex non-Euclidean random objects that are elements of general metric spaces. In such spaces, only pairwise distances between the data objects are available and a strong limitation is that one cannot carry out arithmetic operations due to the lack of an algebraic structure. We combat this complexity by a generalized notion of mean trajectory taking values in the object space. For this, we adopt pointwise Fréchet means and then construct pointwise distance trajectories between the individual time courses and the estimated Fréchet mean trajectory, thus representing the time-varying objects and networks by functional data. Functional principal component analysis of these distance trajectories can reveal interesting features of dynamic networks and object time courses and is useful for downstream analysis. Our approach also makes it possible to study the empirical dynamics of time-varying objects, including dynamic regression to the mean or explosive behavior over time. We demonstrate desirable asymptotic properties of sample based estimators for suitable population targets under mild assumptions. The utility of the proposed methodology is illustrated with dynamic networks, time-varying distribution data and longitudinal growth data.

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1. Introduction

Longitudinal or time-varying data consist of repeated observations for each subject at different time points, where one has such observations for a sample of independent subjects. Tools for analyzing both univariate and multivariate longitudinal data have been well studied (Fieuws and Verbeke 2006; Fitzmaurice et al. 2008; Zhou, Huang, and Carroll 2008; Berrendero, Justel, and Svarc 2011; Xiang, Qiu, and Pu 2013; Verbeke et al. 2014). When such data are scalar or Euclidean vectors and densely measured in time, they can be analyzed as functional data (Rice 2004; Guo 2004; Yang et al. 2007), where functional data analysis (FDA) provides a flexible nonparametric framework with a well-established toolbox. This popular methodology requires a vector space structure of the observed data and is therefore restricted to the case where the measurements at each fixed time are scalars or Euclidean vectors (Ramsay and Silverman 2005; Ferraty, Mas, and Vieu 2007; Horvath and Kokoszka 2012; Hsing and Eubank 2015; Wang, Chiou, and Müller 2016).

Time-varying object data are becoming more frequent and are encountered in time-varying social networks, traffic networks that change over time, brain networks between hubs that evolve with age, and many other settings (Nie, Wang, and Cao 2017). While such data have similarities with densely measured functional data, the observations at each time point are neither scalars nor vectors as in classical FDA, but instead take values in a general metric space. A major challenge is that in such

spaces typical vector space operations such as addition, scalar multiplication or inner products are not defined. In general metric spaces, the only information available are pairwise distances between the random objects at each observation time, and therefore the tools of functional and longitudinal data analysis are not directly applicable. We develop here a simple method that bypasses these challenges by focusing on distances as outcomes.

Models for time courses of non-Euclidean objects have been developed for shape evolution as a continuous diffeomorphic deformation of a baseline shape over time (Durrleman et al. 2013) and for the analysis of longitudinal data taking values in smooth Riemannian manifolds (Muralidharan and Fletcher 2012; Schiratti et al. 2015; Anirudh et al. 2017; Dai and Müller 2018). These methods exploit the local Euclidean nature of Riemannian manifolds but are not applicable for the analysis of data objects in more general metric spaces that do not have a natural Riemannian geometry. For this general case, a fairly complex and challenging methodology has been developed in Dubey and Müller (2019b). We propose here a more straightforward approach that allows us to cut through the challenges posed by longitudinal metric space-valued data by focusing on the distance of time-varying random objects from the mean trajectory and thereby reducing such data to classical functional data.

A related topic is the analysis of longitudinal functional data, where the observations at each time point are functional

rather than scalar. For this scenario, previous approaches (Chen, Delicado, and Müller 2017) have used a tensor product representation of the function-valued stochastic process, an approach for which an underlying Hilbert space is essential and that cannot be directly extended to non-Hilbertian data that we consider here. While complex longitudinal network data have been extensively studied (Huisman and Snijders 2003; Snijders 2005; Kossinets and Watts 2006), these efforts have been directed specifically toward studying dynamics of evolution of a single network with a variety of network effects and are not applicable to the study of the dynamics of a sample of network trajectories or longitudinal trajectories of other general data objects.

Our goal in this article is to provide a straightforward methodology for analyzing *functional object data*, that is, time-varying random objects including dynamic networks that live in general metric spaces. Converting such data to functional data makes it possible to tap the rich existing toolbox of FDA, while imposing not more than mild entropy conditions on the underlying object space. We assume that the data objects take values in a totally bounded metric space and that the random object trajectories are fully observed. Since many dynamic developments of interest can be expressed in terms of departure of an observed dynamic process from a baseline process, we use a suitably defined mean trajectory that takes values in the object space as baseline.

For metric space-valued data, Fréchet developed a generalization of the usual population and sample means (Fréchet 1948), which then gives rise to a generalization of the notion of variance for object data, quantifying the variation of such data around the Fréchet mean. It is thus natural to take as mean trajectory of the object functional data the trajectory consisting of the pointwise Fréchet means, obtained at each time point. To study individual deviations of each time course from this mean trajectory, we first construct squared distance trajectories of the individual object functions from the Fréchet mean trajectory. These squared distance trajectories, which we refer to as subject-specific Fréchet variance trajectories, are scalar valued functional data in contrast to the object trajectories themselves, and therefore can be subjected to the standard tools that have been developed for FDA, including the highly successful functional principal component analysis (FPCA; Kleffe 1973; Hall and Hosseini-Nasab 2006; Li and Guan 2014; Chen and Lei 2015; Lin, Wang, and Cao 2016).

One major obstacle in working with subject-specific Fréchet variance trajectories, which makes it difficult to directly apply functional data methods, is that the population Fréchet mean trajectory is not known and has to be estimated from the data. The squared distance trajectories which are used for the analysis are then the squared distances of the individual time courses from the sample Fréchet mean trajectory. This makes the subject-specific Fréchet variance trajectories dependent and so they cannot be treated as independent observations of random trajectories, which is essential for the application of the usual FDA tools. By imposing mild assumptions on the entropy of the underlying metric space and on the continuity of the random object trajectories, we are able to overcome this problem and to establish desirable asymptotic properties of the

estimators of suitably defined population targets, including rates of convergence.

As we demonstrate in various applications, FPCA of the subject-specific Fréchet variance trajectories can lead to interesting insights regarding the behavior of the object trajectories. Clustering is often a useful first step for exploratory data analysis, aiming to identify homogeneous subgroups and patterns that have some meaningful interpretation for the researcher. Functional data are inherently infinite-dimensional and a probability density generally does not exist, which contributes to the challenge of clustering functional data (Tarpey and Kinateder 2003; Chiou and Li 2007; Jacques and Preda 2014; Ciollaro, Genovese, and Wang 2016; Suarez and Ghosal 2016). An additional difficulty arises when the observations at each time point are not in a vector space. Eigenfunctions of the covariance surface of the Fréchet variance functions can nevertheless pinpoint predominant modes of variation of the individual Fréchet variance trajectories around the average Fréchet variance trajectory and projection scores of the subject-specific Fréchet variance trajectories along these eigenfunctions can reveal inherent clustering within the object trajectories, as we will illustrate in our data applications.

Identifying extremes and potential outliers is challenging for functional data because the observations at a given time point itself may not be unusual in their value while the overall shape of an observed curve may be very different from that of the bulk of curves. Object time courses are even more intractable. Statistical data depth is a concept introduced to measure the “centrality” or “outlyingness” of an observation within a given dataset or an underlying distribution and this concept has been extended to functional data in recent years (López-Pintado and Romo 2009; Nagy, Gijbels, and Hlubinka 2017; Agostinelli 2018; Nagy and Ferraty 2019), where it has been used widely for the detection of extremes and potential outliers in functional data (Febrero, Galeano, and González-Manteiga 2008; Romano and Mateu 2013; Arribas-Gil and Romo 2014; Ren, Chen, and Zou 2017).

An important aspect of our analysis is that conclusions about the behavior of object time courses are drawn based on their squared distances from the Fréchet mean trajectory. The Fréchet mean trajectory is a representative for the most central point for a sample of object functions and the subject-specific Fréchet variance time courses carry information about the deviations of individual trajectories from the “central” trajectory, which leads to a central-outward ordering for the sample trajectories. We show that principal component projection scores of the subject-specific Fréchet variance time courses along eigenfunctions are useful for visualization of the longitudinal object data and also for the detection of extremes. Another aspect of interest is the dynamics of the evolving object trajectories, especially whether they tend to move closer to the mean function as time progresses, so that a faraway trajectory will tend to be drawn toward the center (centripetality) or will move further away from the center (centrifugality), as time progresses.

The article is organized as follows: In Section 2, we introduce our framework and define the population targets and the corresponding sample-based estimators. The theoretical properties of the estimators are established in Section 3. This is followed

by data illustrations in Section 4, where we apply the proposed method for the longitudinal network generated by the Chicago Divvy bike data for the years 2014–2017, for the longitudinal annual fertility data for 26 countries over 34 calendar years from 1976 to 2009 and for time-varying shape data using the Zürich longitudinal growth study. We also demonstrate the proposed quantification of the underlying dynamics of the observed processes. Simulation results for a sample of time-varying networks are presented in Section 5, followed by a discussion in Section 6. Auxiliary results and proofs can be found in the online supplement.

2. Preliminaries and Estimation

We consider an object space (Ω, d) that is a totally separable bounded metric space and an Ω -valued stochastic process $\{X(t)\}_{t \in [0,1]}$, alternatively referred to as X for ease of notation, and assume that one observes a sample of random object trajectories $X_1, X_2, \dots, X_n$, which are independently and identically distributed copies of the random process X with respect to an underlying probability measure P . For each subject-specific trajectory X_i , we aim to quantify its deviation from a baseline object function, which can be thought of as the mean or typical population trajectory. A natural baseline for real-valued functional data is the mean function. For more general object-valued trajectories, we propose to use the population Fréchet mean trajectory as baseline function, defined as the pointwise Fréchet mean function, where for given $t \in [0, 1]$, the population and sample Fréchet mean trajectories at t are defined as

$$\begin{aligned}\mu(t) &= \operatorname{argmin}_{\omega \in \Omega} E(d^2(X(t), \omega)), \\ \hat{\mu}(t) &= \operatorname{argmin}_{\omega \in \Omega} \frac{1}{n} \sum_{i=1}^n d^2(X_i(t), \omega),\end{aligned}\quad (1)$$

respectively. Here, we assume that for all $t \in [0, 1]$ these minimizers exist and are unique. While the existence and uniqueness of Fréchet means is not guaranteed in general spaces (Bhattacharya and Patrangenaru 2003), for the case of Hadamard spaces, which have globally nonpositive curvature, Fréchet means as defined in Equation (1) exist and are unique (Sturm 2003). For positively curved spaces, see Ahidar-Coutrix, Le Gouic, and Paris (2020).

The target functions for our analysis then ideally would be the functions

$$V_i^*(t) = d^2(X_i(t), \mu(t)), \quad t \in [0, 1], \quad (2)$$

which correspond to the pointwise squared distance functions of the subject trajectories X_i from the population Fréchet mean function $\mu = \mu(t)$ for the subject trajectories X_i . These can be characterized as the subject-specific oracle Fréchet variance trajectories. They are however unavailable, since the population Fréchet mean trajectory μ is unknown and needs to be estimated from the data. From Equation (2) one obtains the data-based version

$$V_i(t) = d^2(X_i(t), \hat{\mu}(t)), \quad t \in [0, 1], \quad (3)$$

where $\hat{\mu}$ is as in Equation (1) and we refer to the $V_i = V_i(t)$ as the sample Fréchet variance trajectories and write $V(t) =$

$d^2(X(t), \hat{\mu}(t))$ for the generic version. Since they all depend on $\hat{\mu}(t)$, the sample Fréchet variance trajectories are dependent and cannot be treated as independent realizations of a stochastic process, which is the standard framework for FDA, thus posing a challenge for theory.

Suppose for the moment that we have available an iid sample of oracle Fréchet variance trajectories V_i^* , with generic version denoted by V^* . Then a typical dimension reduction step in FDA is to apply FPCA, which facilitates the conversion of the functional data V_i^* to a countable sequence of uncorrelated random variables, the functional principal components (FPCs), where the sequence of FPCs is often truncated at a finite-dimensional random vector to achieve dimension reduction. FPCA is based on using the eigenfunctions of the auto-covariance operator of the process V^* . This is an integral operator, a trace class and moreover a compact Hilbert Schmidt operator (Hsing and Eubank 2015) that has the population Fréchet covariance surface C as its kernel, where

$$\begin{aligned}C(s, t) &= E(d^2(X(s), \mu(s))d^2(X(t), \mu(t))) \\ &\quad - E(d^2(X(s), \mu(s))) E(d^2(X(t), \mu(t))).\end{aligned}\quad (4)$$

The eigenvalues of the auto-covariance operator are nonnegative as the covariance surface is symmetric and nonnegative definite. By Mercer's theorem,

$$C(s, t) = \sum_{j=1}^{\infty} \lambda_j \phi_j(s) \phi_j(t), \quad s, t \in [0, 1],$$

with uniform convergence, where the λ_j are the eigenvalues of the covariance operator, ordered in decreasing order, and $\phi_j(\cdot)$ are the corresponding orthonormal eigenfunctions.

This leads to the Karhunen–Loève expansion of the oracle Fréchet variance trajectories,

$$V_i^*(t) = v^*(t) + \sum_{j=1}^{\infty} B_{ij} \phi_j(t), \quad (5)$$

with L^2 convergence. Here, v^* is the mean function of the subjectwise Fréchet variance functions, $v^*(t) = E(d^2(X(t), \mu(t)))$, the population Fréchet variance function. The B_{ij} are the FPCs, which are uncorrelated across j with $E(B_{ij}) = 0$, $\operatorname{var}(B_{ij}) = \lambda_j$ and $B_{ij} = \int (V_i^*(t) - v^*(t)) \phi_j(t) dt$. If μ is known, according to Equation (4), then the oracle estimator of the Fréchet covariance surface is

$$\tilde{C}(s, t) = \frac{1}{n} \sum_{i=1}^n V_i^*(s) V_i^*(t) - \frac{1}{n} \sum_{i=1}^n V_i^*(s) \frac{1}{n} \sum_{i=1}^n V_i^*(t). \quad (6)$$

Under mild assumptions on the functional trajectories V_i^* , standard asymptotic theory from FDA shows that this estimator has desirable asymptotic properties and converges to the true covariance surface C (4) (Hall and Hosseini-Nasab 2006).

As the population Fréchet mean function μ in reality is however unknown, we need to replace it by the sample-based estimator $\hat{\mu}$ of the Fréchet covariance surface,

$$\hat{C}(s, t) = \frac{1}{n} \sum_{i=1}^n V_i(s) V_i(t) - \frac{1}{n} \sum_{i=1}^n V_i(s) \frac{1}{n} \sum_{i=1}^n V_i(t), \quad (7)$$

using the data-based distance processes $V_i(t)$ Equation (3) that depend on estimates $\hat{\mu}$. We show in Section 3 that $\hat{C}$ is asymptotically close to the oracle estimator of the Fréchet covariance surface $\tilde{C}$ under mild regularity conditions on the metric space and the object functions and therefore has desirable asymptotic properties as an estimator of the population Fréchet covariance surface.

Estimates of eigenvalues and eigenfunctions are obtained as the empirical eigenvalues and eigenfunctions of the integral covariance operator with covariance kernel $\hat{C}$, and will be denoted by $\hat{\lambda}_j$ and $\hat{\phi}_j$, ordered in decreasing order of the eigenvalues. Eigenfunctions ϕ_j can be interpreted as coordinate directions, thereby providing the basis for principal modes of variation of the subject-specific oracle Fréchet variance trajectories around the population Fréchet variance trajectory. Modes of variation (Castro, Lawton, and Sylvestre 1986; Lila and Aston 2019) are useful to quantify the departure of a random object trajectory from the Fréchet mean function. The eigenfunctions can be viewed as *modes of outlyingness* of the subject-specific trajectories. The estimates of the projection scores B_{ij} of the i th oracle Fréchet variance trajectory on the j th eigenfunction are given by

$$\hat{B}_{ij} = \int_0^1 \left(V_i(t) - \frac{1}{n} \sum_{k=1}^n V_k(t) \right) \hat{\phi}_j(t) dt. \quad (8)$$

We show in Section 3 that under regularity assumptions, the $\hat{B}_{ij}$ are asymptotically close to the B_{ij} in Equation (5). These scores are useful for visualizing common traits in object trajectories and for detecting extremes, homogeneous subgroups or clusters in the data.

3. Theory

We establish asymptotic properties of the empirical estimators of the population targets as described in Section 2, assuming that the realizations of the object valued random process $X(t)$, $t \in [0, 1]$, have continuous sample paths almost surely, take values in a totally bounded metric space (Ω, d) and the following conditions are satisfied:

- (A1) The objects $\mu(s)$ and $\hat{\mu}(s)$ exist and are unique, the latter almost surely, for each $s \in [0, 1]$. Additionally for any $\epsilon > 0$,

$$\inf_{s \in [0, 1]} \inf_{\omega \in \Omega: d(\omega, \mu(s)) > \epsilon} E(d^2(X(s), \omega)) - E(d^2(X(s), \mu(s))) > 0$$

and there exists a $\tau = \tau(\epsilon) > 0$ such that

$$\lim_{n \rightarrow \infty} P \left(\inf_{s \in [0, 1]} \inf_{\omega \in \Omega: d(\omega, \hat{\mu}(s)) > \epsilon} \frac{1}{n} \sum_{i=1}^n \{d^2(X_i(s), \omega) - d^2(X_i(s), \hat{\mu}(s))\} \geq \tau(\epsilon) \right) = 1.$$

- (A2) There exists $\rho > 0$, $D > 0$ and $\beta > 1$ such that

$$\inf_{s \in [0, 1]} \inf_{\omega \in \Omega: d(\omega, \mu(s)) < \rho} \{E(d^2(X(s), \omega)) - E(d^2(X(s), \mu(s))) - Dd^\beta(\omega, \mu(s))\} \geq 0.$$

- (A3) For some $0 < \alpha \leq 1$, the random function $X(\cdot)$ defined on $[0, 1]$ and taking values in Ω , where we denote the space of all such functions as $\Omega^{[0, 1]}$, is α -Hölder continuous, that is, for nonnegative $G: \Omega^{[0, 1]} \rightarrow \mathbb{R}^+$ with $E(G(X)^2) < \infty$, it holds almost surely,

$$d(X(s), X(t)) \leq G(X)|s - t|^\alpha.$$

- (A4) For $I(\delta) = \int_0^1 \sup_{s \in [0, 1]} \sqrt{\log N(\epsilon \delta, B_\delta(\mu(s)), d)} d\epsilon$ it holds that $I(\delta) = O(1)$ as $\delta \rightarrow 0$. Here $B_\delta(\mu(s)) = \{\omega \in \Omega : d(\omega, \mu(s)) < \delta\}$ is the δ -ball around $\mu(s)$ and $N(\gamma, B_\delta(\mu(s)), d)$ is the covering number, that is, the minimum number of balls of radius γ required to cover $B_\delta(\mu(s))$ (Van der Vaart and Wellner 1996).

Assumption (A1) guarantees uniform convergence of the sample Fréchet mean trajectory to its population target as it implies $\sup_{s \in [0, 1]} d(\hat{\mu}(s), \mu(s)) = o_P(1)$ (Dubey and Müller 2019b); for convenience, we state this result as Lemma 3 in the supplement without proof. Measurability issues of the sample Fréchet mean function can be dealt with in a similar fashion as M -estimators in general by considering outer probability measures; for more detailed discussion of the measurability issues see sections 1.2, 1.3 and 1.7 of Van der Vaart and Wellner (1996). Assumptions of type (A2) are standard for M -estimators and characterize the local curvature of the target function to be minimized near the minimum; this curvature is characterized by β , which features in the resulting rate of convergence. Lemma 5 in section A.3 of the supplement provides the rate of convergence of the Fréchet mean function $\hat{\mu}(s)$, and corrects an algebraic error in Theorem 3 in Dubey and Müller (2019b). When (Ω, d) is a Hadamard space, β takes the value 2 (Sturm 2003) for any probability measure on Ω , and therefore assumptions (A1) and (A2) are satisfied.

Assumptions (A3) and (A4) are required for measuring the size of the space of object functions and imply an entropy condition on the object function space, which then leads to uniform convergence of the plug-in estimator of the Fréchet covariance surface $\tilde{C}$ in Equation (6) given by $\hat{C}$ (7) at a fast rate. In (A3), we assume that the rate of Hölder continuity of the random object trajectories is fixed, with the Hölder constant having a finite second moment, which means that $E \left\{ \left(\sup_{s \neq t, s, t \in [0, 1]} d(X(s), X(t)) / |s - t|^\alpha \right)^2 \right\}$ is finite. This assumption is a mild smoothness assumption satisfied for certain values of α by many common Euclidean-valued random processes, including the Wiener process for $\alpha = \frac{1}{2}$.

Assumption (A3) together with the curvature condition in (A2) implies Hölder continuity of the Fréchet mean function $\mu(\cdot)$. For details, we refer to the proof of Lemma 4 in section A.3 of the supplement. Assumption (A4) is a bound on the covering number of the object metric space and is satisfied by several commonly encountered random objects, including random probability distributions equipped with the 2-Wasserstein metric, covariance matrices of fixed dimension and graph Laplacians of networks with fixed number of nodes (Dubey and Müller 2019a; Petersen and Müller 2019). We provide a proof and further discussion on this in section A.5 of the supplement, where we show that the space of univariate distributions with the 2-Wasserstein metric and the space of graph Laplacians with the Frobenius metric satisfy assumptions (A1)–(A4).

A consequence of Lemma 2 in section A.2 of the supplement is that $\sqrt{n}(\hat{C}(s, t) - C(s, t))$ converges weakly to a Gaussian process limit, which then leads to

$$\sqrt{n} \sup_{s, t \in [0, 1]} |\hat{C}(s, t) - C(s, t)| = O_P(1)$$

by an application of the uniform mapping theorem.

Theorem 1 (Weak convergence of $\hat{C}$). Under assumptions (A1)–(A4), the process $\sqrt{n}(\hat{C}(s, t) - C(s, t))$ converges weakly to a zero-mean Gaussian process limit with covariance function $\mathcal{R}$, which for $(s_1, t_1), (s_2, t_2) \in [0, 1]^2$ is given by

$$\begin{aligned} \mathcal{R}((s_1, t_1), (s_2, t_2)) \\ &= \text{cov}((V^*(s_1) - v^*(s_1))(V^*(t_1) - v^*(t_1)), \\ &\quad (V^*(s_2) - v^*(s_2))(V^*(t_2) - v^*(t_2))), \\ &\text{where} \quad V^*(s) = d^2(X(s), \mu(s)), \quad v^*(s) = E(V^*(s)). \end{aligned}$$

This weak convergence result provides the major justification that observed processes V may be used for the proposed FPCA instead of the oracle processes V^* when the mean function has to be estimated, as will invariably be the case in practical applications. Uniform convergence and rates of convergence of $|\lambda_j - \hat{\lambda}_j|$ and $\sup_{s \in [0, 1]} |\hat{\phi}_j(s) - \phi_j(s)|$ then follow from Theorem 1 by standard perturbation results along the lines of the Davis-Kahan theorem, for example, Lemma 4.3 of Bosq (2000). Our next result provides a quantification of the asymptotic closeness of the sample-based estimators of the FPCs and the oracle FPCs and requires the additional assumption

(A6) For each $j \geq 1$, the eigenvalue λ_j , as defined in Section 2, has multiplicity 1, that is, it holds that $\delta_j > 0$ where $\delta_j = \min_{1 \leq l \leq j} (\lambda_l - \lambda_{l+1})$.

Theorem 2. Under assumptions (A1)–(A6), we have

$$\begin{aligned} |\hat{\lambda}_j - \lambda_j| &= O_P\left(\frac{1}{\sqrt{n}}\right), \\ \sup_{s \in [0, 1]} |\hat{\phi}_j(s) - \phi_j(s)| &= O_P\left(\frac{1}{\delta_j \sqrt{n}}\right), \\ |\hat{B}_{ij} - B_{ij}| &= O_P\left(\frac{1}{\delta_j \sqrt{n}}\right) + O_P\left(\left(\frac{\sqrt{\log n}}{n}\right)^{\frac{1}{2(\beta-1)}}\right). \end{aligned}$$

Examples of object spaces that satisfy the assumptions include graph Laplacians of connected, undirected and simple graphs corresponding to networks of fixed dimension equipped with the Frobenius metric (Dubey and Müller 2019a), as well as univariate probability distributions equipped with the 2-Wasserstein metric and correlation matrices of a fixed dimension equipped with the Frobenius metric (Petersen and Müller 2019). For all of the above examples, one has $\beta = 2$ in assumption (A2). For a detailed discussion, see Section A.5 in the supplement.

4. Data Illustrations

4.1. Chicago Divvy Bike Data

The Chicago Divvy bicycle sharing system makes historical bike trip data publicly available at <https://www.divvybikes.com/system-data>. The dataset includes trip start and end dates and times, duration, start and end locations and anonymized rider data. The bike trip details are recorded at a resolution of seconds in time and include trips between 580 bike stations in Chicago and two adjacent suburbs. We used a cleaned version of the trip records of duration one hour or less between 2013 to 2017, which are available at <https://www.kaggle.com/yingwurenjian/chicago-divvy-bicycle-sharing-data>.

These data were also analyzed by Gervini and Khanal (2019), who applied a functional version of point processes. Studying patterns in the daily evolution of the number of bike rides between various bike stations can provide insights into the Divvy bike sharing system and patterns of transport in the city. We study time-varying networks that are defined by the number of bike trips between stations of interest and the evolution of these networks during a typical day. We constructed samples of time-varying networks observed over a sample of 1457 days in the years 2013–2017.

We focus our analysis on data pertaining to the area east of Greektown, south of Wrigley field and north of Chinatown containing the Lakefront trails, the Navy pier and many other popular destinations. We considered 112 popular bike stations in this region and each day was broken into 20 min intervals. On each of these intervals we constructed a network with 112 nodes, each one corresponding to one of the bike stations and edge weights representing the number of recorded bike trips between the pairs of stations that define the edges of the network within the 20 min interval. This generates a time-varying network for each of the 1457 days in the years 2013–2017 for which complete records are available. The time points where the network is sampled over the course of each day were chosen as the mid-points of the 20 min intervals of a day. The observations at each time point correspond to a 112-dimensional graph Laplacian that characterizes the network between the 112 bike stations of interest for that particular 20 min interval. For a network with r nodes, the adjacency matrix is a $r \times r$ matrix A , where the $(i, j)^{\text{th}}$ entry a_{ij} represents the edge weight between nodes i and j . The graph Laplacians L are given by $L = D - A$, where D is the degree matrix, the off diagonal entries of which are zero, with diagonal entries $d_{ii} = \sum_{j=1}^r a_{ij}$. The graph Laplacians determine the network uniquely.

We used the Frobenius metric as a distance measure between graph Laplacians. The sample Fréchet mean trajectory at a particular time point therefore is the sample average of the graph Laplacians of 1457 networks corresponding to different days for that time point. We then obtained the Fréchet variance trajectories for each day, which for a given day and time point correspond to the squared Frobenius distance between the graph Laplacian and the Fréchet mean graph Laplacian, and then applied FPCA for the resulting 1457 Fréchet variance trajectories.

The mean Fréchet variance trajectory of the daily graph Laplacians for the Divvy bike trip networks as a function of

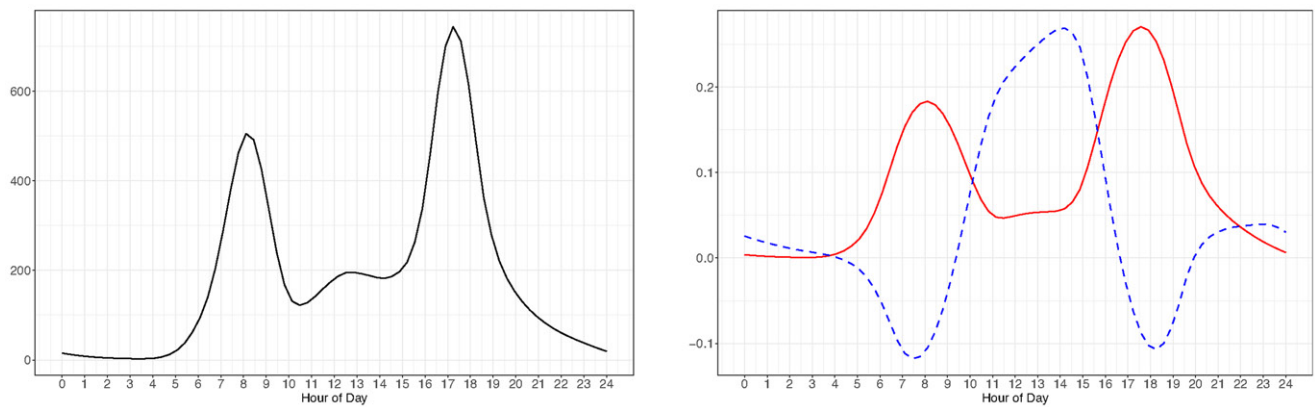


Figure 1. Sample mean function (left plot) and eigenfunctions for the FPCA (right plot) of the squared distance trajectories at 20 min intervals of graph Laplacians of daily Divvy bike trip networks in Chicago. In the right plot, the solid red line corresponds to the first eigenfunction, which explains 90.40% of variability in the trajectories and the dashed blue line to the second eigenfunction, which explains 7.28% of the variability.

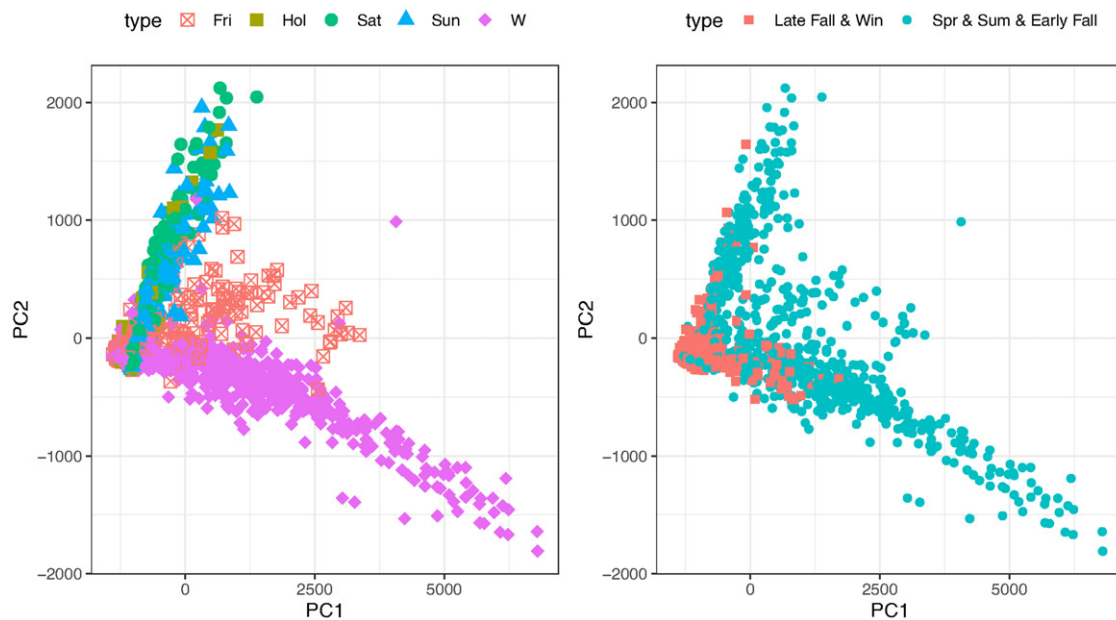


Figure 2. Pairwise plots of the first two FPC scores, distinguished by day of the week (left plot) and by season (right plot). In the left plot, “W” stands for regular Mondays to Thursdays, “Fri” for Fridays, “Sat” for Saturdays, “Sun” for Sundays, and “Hol” for special holidays. In the right plot, “Late Fall & Win” includes months between November and March and “Spr & Sum & Early Fall” includes months from April to October.

the time within the day, which quantifies the average squared deviation from the mean trajectory, is shown in the left plot of Figure 1. The peaks are at 9 a.m. with elevated mean variation between 7 a.m. to 10 a.m. and at 6 p.m. with elevated levels between 4 p.m. and 7 p.m., which reflect morning and late afternoon and early evening commuting surges, where the network variation is seen to be highest.

The predominant directions of variation of the daily Fréchet variance trajectories around the Fréchet mean function are visualized by the first two eigenfunctions in the right plot of Figure 1. The first two functional principal component scores explain about 97.7% of the variation in the Fréchet variance trajectories. The first eigenfunction reflects increased variability around the peaks of the Fréchet variance function that is shown in Figure 1. The peaks are between between 7 a.m. to 10 a.m. and 4 p.m. to 7 p.m., which reflect morning and late afternoon and early evening peaks of commute, where the deviations from the mean Fréchet variance function are seen to be largest. The

second eigenfunction reflects a contrast between these peaks and the squared deviation from the mean Fréchet variance function during the time period 11 a.m. to 4 p.m.

Analyzing the FPC scores of the daily Fréchet variance trajectories along the first and second eigenfunctions, Figure 2 reveals several interesting patterns in the daily Fréchet variance trajectories. Weekdays and weekends form distinct clusters. Holidays show patterns similar to weekends. When scrutinizing second versus first FPC scores, an outlying observation is found at August 21, 2017. Researching the background of this day, we found that there was a total solar eclipse, which was in peak view over Chicago at 1:18 p.m. in the afternoon. The Fréchet variance trajectory of this particular day is illustrated in Figure A.1 in Section A.4 of the online supplement. An application of the naive Bayes classifier and the support vector machine on the first two FPC scores of the daily Fréchet variance trajectories with “weekdays” and “weekends and holidays” as the binary response using 75% of the data as training sample

and 25% of the data as test sample gave a low misclassification rate of 6.02% in both cases. The classification result is illustrated in Figure A.2 in the online supplement.

We also performed a FPCA of these Fréchet variance trajectories for weekdays, Fridays and weekends, including special holidays in the same group as weekends, separately for the three cases, with results displayed in Figures A.3 and A.4 in the online supplement. Weekdays and weekends, including special holidays, show clear differences in both mean Fréchet variance functions and eigenfunctions. The Friday pattern can be characterized as “transition” from weekdays to weekends. Seasonal differences can impact bike sharing patterns. In the right plot of Figure 2, we display second versus first FPC scores, differentiated according to two broad seasonal groups. Spring, summer and early fall includes months from April to October that exhibit greater variability than the late fall and winter months of November to March, with further illustrations in Figures A.5 and A.6 in the online supplement.

4.2. Fertility Data

The Human Fertility Database provides cohort fertility data for various countries and calendar years. The data are available

at www.humanfertility.org and facilitate the study of the time evolution and inter-country differences in fertility over a period spanning more than 30 calendar years (Chen, Delicado, and Müller see also 2017). We selected 27 countries with complete fertility records for the time period 1976–2009. For each country and year, the age-specific total live birth counts correspond to histograms of maternal age with bin size one year. These histograms were smoothed (for which we employed local least-squares smoothing using the Hades package available at <https://stat.ucdavis.edu/hades/>) to obtain smooth probability density functions for maternal age, where we consider the age interval [12, 55]. We thus obtain samples of time-varying univariate probability distributions, where the subjects are the countries, the time is calendar years between 1976 and 2009 and the observation at each time point for a specific country is its maternal age distribution, that is, the distribution of ages when females give birth within the age interval [12, 55] for the specified country and calendar year.

Figure 3 displays the evolving densities for Japan, the United States, Portugal, and Bulgaria in some selected years from 1976 to 2010. There are clear differences in the maternal age evolution between countries, but the overall trend is that maternal age increases. This is also reflected in the Fréchet mean densities

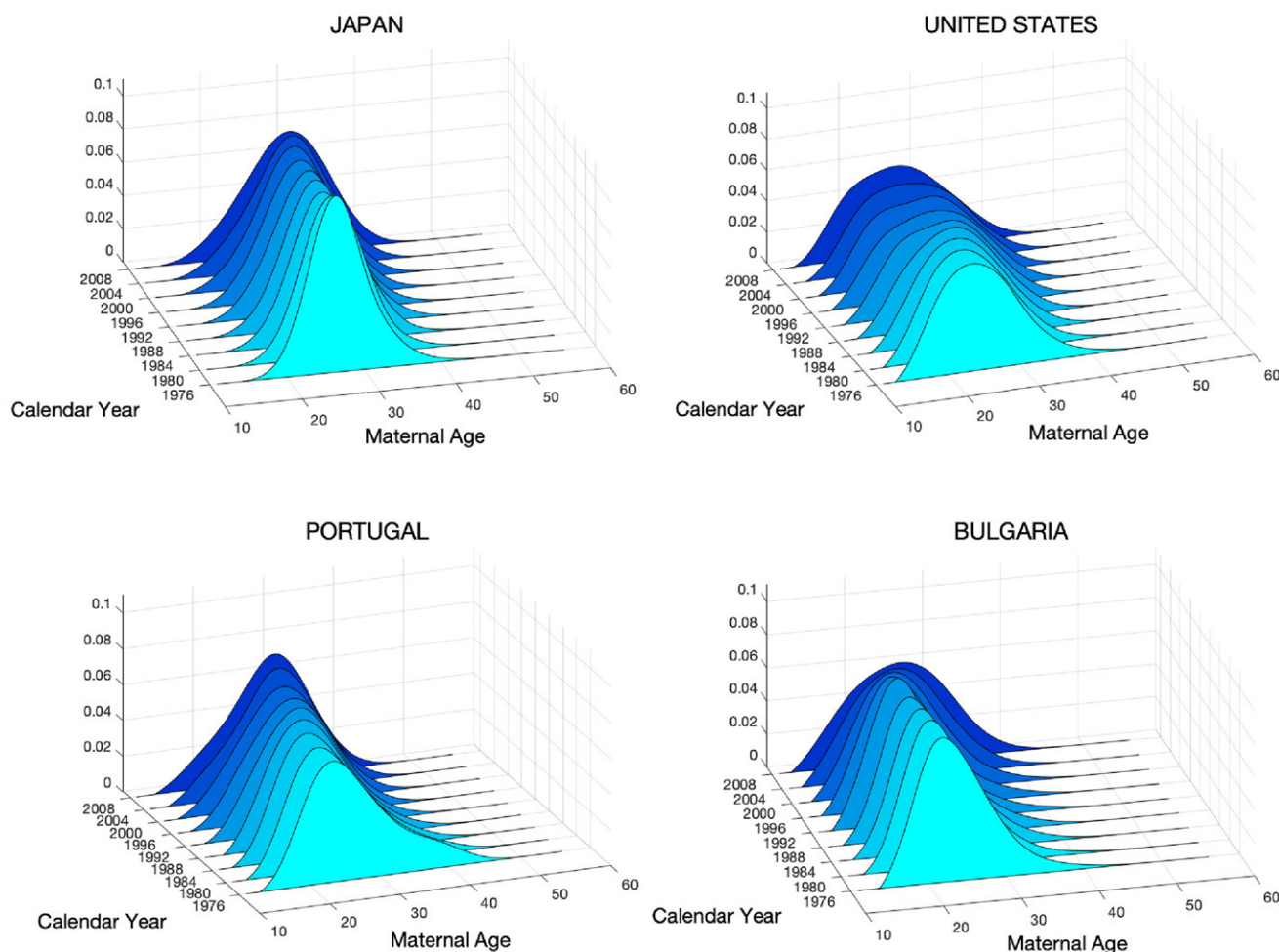


Figure 3. Yearly maternal age distributions represented as density functions for the age interval [12, 55] during the time period 1976–2009 (selected years). The x-axis represents maternal age, the z-axis the density functions and the y-axis calendar years.

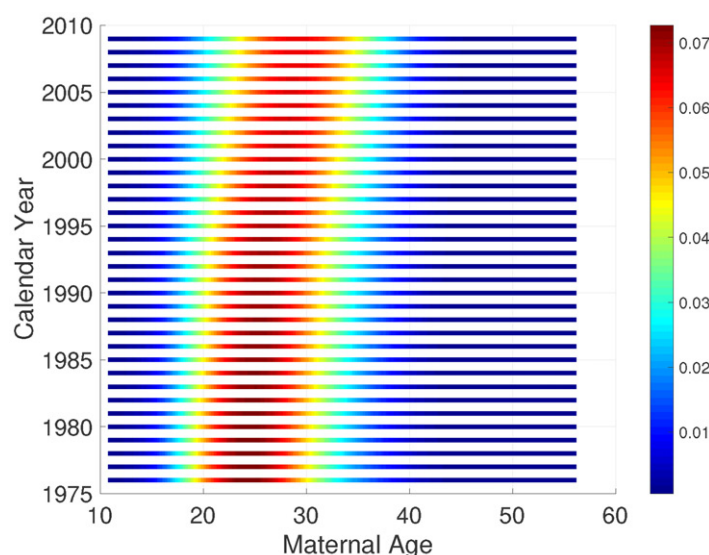


Figure 4. Fréchet mean maternal age distributions represented as a heat plot of density functions for the age interval [12,55] during the time period 1976–2009.

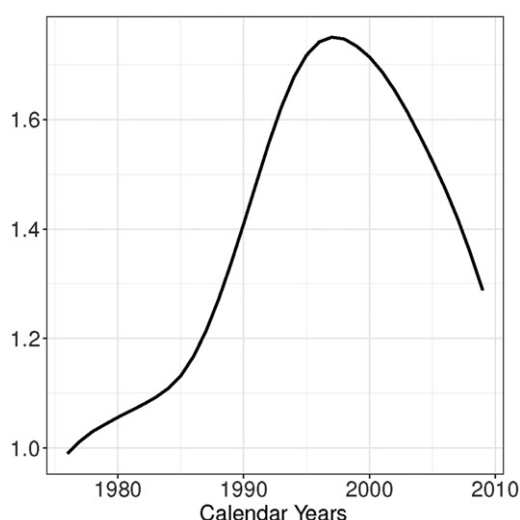


Figure 5. Sample mean function of the Fréchet variance trajectories of the time courses of country-specific maternal age distributions.

in Figure 4, which show a shift in their mode locations toward higher age over the years.

We opt for the 2-Wasserstein metric as the distance between probability distributions, which corresponds to the L^2 distance between their quantile functions, a metric that has proved to be an excellent choice in many applications (Bolstad et al. 2003; Bigot, Cazelles, and Papadakis 2018). The sample Fréchet mean trajectory in a particular year then corresponds to the sample average of the quantile functions of the 27 countries in that year (Dubey and Müller 2019a), which is represented as the corresponding density function. We then obtained the Fréchet variance trajectory for each country, where its value for a given calendar year corresponds to the squared 2-Wasserstein distance between the maternal age distribution of the country to the Fréchet mean age distribution for the specified calendar year.

Finally, we performed an FPCA on the 27 country-specific Fréchet variance trajectories. Figure 5 shows the estimated mean function of the Fréchet variance trajectories, which is a scalar function that corresponds to the sample average of the

Fréchet variance trajectories as a function of calendar year. The predominant directions of variation of the distance trajectories around the Fréchet mean trajectory are captured by the first two eigenfunctions, which are depicted in the left panel of Figure 6. The first and second eigencomponent explain 74.64% and 17.40% of the variation in the distance trajectories. The first eigenfunction is increasing until between 1995 and 2000 and then starts to decrease. This shows increasing deviation of the distance trajectories from the mean Fréchet variance trajectory until right before the new millennium after which these deviations tend to decrease. The second eigenfunction reflects a contrast between the earlier and later parts of the calendar time interval.

The years between 1995 and 2000 mark the critical period when the changes in the mode of maternal age distributions start to take place as displayed in Figure 4. This period of increased activity is also prominent in Figure 5 which shows that the Fréchet variance of maternal age distributions increases in the beginning, reaches a peak between 1995 and 2000 and then starts to decrease. This might be attributed to increasing numbers of women opting for higher education and participating in the labor force in some of the countries included in the dataset over the time interval where the data have been collected. Another likely factor are advanced birth control measures in the past few decades, which led to changes in the maternal age distribution early on for some countries, while these changes were delayed for some other countries, leading to increased discrepancies between countries from 1995 to 2000, which stabilized later as countries moved closer to the mean behavior.

The FPC scores of the trajectories along predominant directions of variation not only reveal interesting patterns but also aid in identifying extremes, which is tricky for the case of time-varying probability distributions. Plotting second versus first FPC scores (right plot of Figure 6), one finds that Bulgaria shows large deviations from the Fréchet mean variance function trajectory along both the first and second eigenfunctions. The Czech Republic has the highest second FPC, indicating a large contrast in the distance from the Fréchet mean trajectory

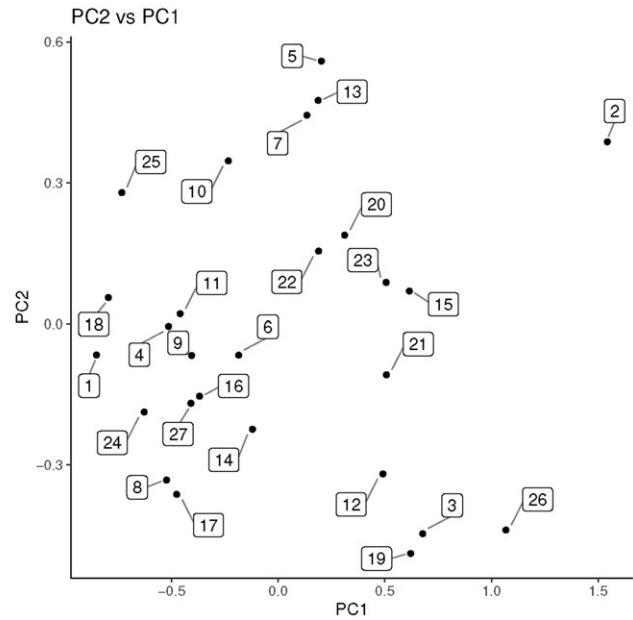
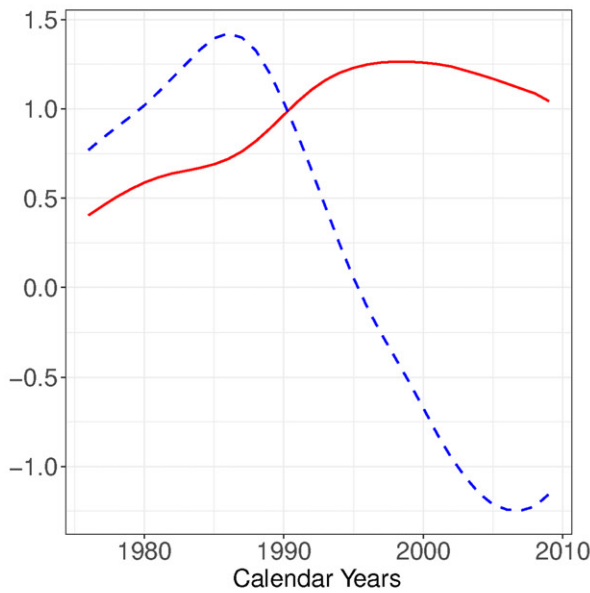


Figure 6. Left panel: First (solid red) and second (blue dashed) eigenfunction of the country-specific Fréchet variance trajectories, which explain 74.64% and 17.40% of variability, respectively. Right panel: Second versus first FPC scores of the country-specific Fréchet variance trajectories. Here 1 denotes Austria; 2 Bulgaria; 3 Belarus; 4 Canada; 5 Czech Republic; 6 Estonia; 7 Finland; 8 Germany; 9 France; 10 Hungary; 11 Israel; 12 Italy; 13 Japan; 14 Lithuania; 15 the Netherlands; 16 Norway; 17 Poland; 18 Portugal; 19 Russia; 20 Slovakia; 21 Spain; 22 Sweden; 23 Switzerland; 24 the United Kingdom; 25 Taiwan; 26 Ukraine; and 27 denotes the United States.

between early and later years. Portugal and Austria have negative first FPCs and small second FPCs, which means that their deviation from the Fréchet mean trajectory is less than the average deviation over the calendar time interval.

4.3. Empirical Dynamics

When using FDA for analyzing real-valued longitudinal data, a common assumption is that the data are generated by an underlying smooth and square integrable stochastic process. The derivatives of the trajectories of such processes are often the key to understanding the underlying dynamics (Ramsay 2000; Mas and Pumo 2009; Ramsay et al. 2007). Empirical dynamics (Müller and Yao 2010) is a systematic approach to assess the underlying dynamics of longitudinally observed data. It is based on the decomposition

$$Y^{(1)}(t) - \mu^{(1)}(t) = \beta(t)\{Y(t) - \mu(t)\} + Z(t), \quad (9)$$

where $Y(\cdot)$ is the underlying stochastic process, $\mu(t) = E(Y(t))$, $Y^{(1)}(\cdot)$ and $\mu^{(1)}(\cdot)$ are the derivatives of $Y(\cdot)$ and $\mu(\cdot)$, $\beta(\cdot)$ is a smooth time-varying linear coefficient function and $Z(\cdot)$ is a random drift process. For Gaussian processes this decomposition can be easily derived and then Z and Y are independent with $E(Z(t)) = 0$, leading to

$$E[Y^{(1)}(t) - \mu^{(1)}(t) | Y(t)] = \beta(t)\{Y(t) - \mu(t)\}. \quad (10)$$

When one has non-Gaussian processes, such as the distance processes $V^*(2)$, the above decompositions provide useful approximations and can be interpreted in a least-square sense, where the coefficient of determination

$$R^2(t) = 1 - \frac{\text{var}\{Z(t)\}}{\text{var}\{Y^{(1)}(t)\}}$$

indicates the fraction of variance explained by the empirical dynamics approximation. On pertinent subdomains where $R^2(t)$ is relatively large, the dynamics of the trajectories are determined to a greater extent by the linear model in Equation (10), while otherwise the random drift process Z becomes the driving factor instead of the dynamic equation. The time-varying function $\beta(t)$ summarizes the characteristics of the dynamics of the underlying process. If $\beta(t) < 0$, one observes *centripetality* or *dynamic regression to the mean*, that is, a trajectory which is away from the mean function tends to move closer toward the mean function as time progresses. If on the other hand $\beta(t) > 0$, one has *centrifugality* or *dynamic explosive behavior*, since deviations from the mean at time t tend to increase beyond t .

Noting that it is infeasible to define derivatives for trajectories of random objects since the notion of derivative for metric space data is not defined, we apply empirical dynamics instead to the real-valued Fréchet variance trajectories $V^*(2)$,

$$\frac{d}{dt}(V^*(t) - EV^*(t)) = \beta(t)(V^*(t) - EV^*(t)) + Z(t). \quad (11)$$

We note that centripetality and centrifugality relate to the mean trajectory $EV^*(t)$, that is, the mean Fréchet variance function; we quantify the dynamics of the variation relative to this function. For estimation of the time-varying coefficient $\beta(\cdot)$ and $R^2(t)$, we adopt a plug-in estimation procedure (Liu and Müller 2009; Sentürk and Müller 2010).

We implemented the empirical dynamics model (11) for the sample of time-varying maternal age distributions with the R package *fdapace*. Figure 7 illustrates the estimated slope function $\hat{\beta}(t)$ and coefficient of determination function $\hat{R}^2(t)$, where the latter indicates that the dynamics of the maternal age distribution trajectories can be explained by the first order differential equation in the period before 1995 and after 2000,

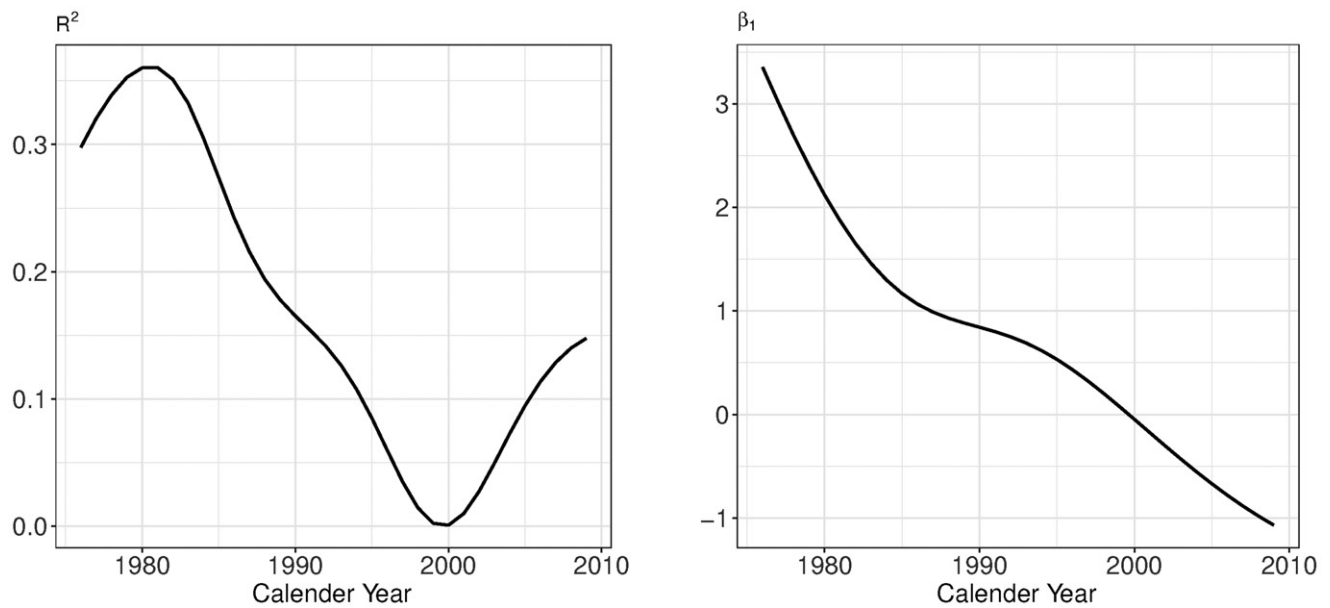


Figure 7. Smooth estimate of the coefficient of determination $R^2(t)$ (left) and the varying coefficient function $\beta(t)$ (right), capturing empirical dynamics of the distance trajectories of the fertility data.

but not in between. The slope function is positive before 2000 and negative after 2000, indicating centrifugality of the maternal age distributions before 2000 and centripetality after 2000, so that in more recent years there is a tendency for the fertility distributions to become more similar in the sense that their variation around the mean distance function is decreasing over calendar time, while the mean distance function itself is also decreasing as seen in Figure 5.

4.4. Zürich Longitudinal Growth Data

Statistical shape analysis is an emerging field of data analysis that constitutes measuring, describing and comparing random shapes (Kendall 1984, 1989; Le and Kendall 1993; Mardia et al. 2005; Patrangenaru and Ellingson 2015; Dryden and Mardia 2016). Often, shapes are determined using a finite set of coordinate points, known as landmark points. As an example of shapes evolving over time, we consider the growth modalities that are obtained from a longitudinal study on human growth and development (Gasser et al. 1984). This study included growth data for 232 Swiss children and was conducted in the University Children's Hospital in Zürich between 1954 and 1978. For each child, we consider trajectories $X_i(t)$, where $X_i(t)$ is a 4×2 matrix which represents four landmarks in two dimensions. The four landmarks are the foot, which is set to be the point (0, 0), the top of the head which has coordinate (0, standing height), the shoulder which is set to have the coordinate (bi-humeral diameter, 0.8 standing height) and the hip which is set at the point (bi-iliacal diameter, leg length). These shape trajectories are observed for the age interval $t \in [0.5, 20]$ years.

The trajectories $X_i(t)$ described above correspond to two-dimensional landmarks that take values in the planar shape space, which can be viewed as configurations on the complex plane. In this space, we adopt the full Procrustes metric, which has been quite successful in the analysis of planar shapes in a

wide variety of applications (Dryden and Mardia 2016). Given centered complex configurations $y = (y_1, y_2, \dots, y_k)$ and $z = (z_1, z_2, \dots, z_k)$, both in $\mathbb{C}^k$, with $y^* 1_k = 0 = z^* 1_k$, the full Procrustes distance between y and z is defined as

$$d_P(y, z) = \left\{ 1 - \frac{y^* z z^* y}{z^* z y^* y} \right\}^{1/2}.$$

Accordingly, we first obtained the sample Fréchet mean trajectory of the random shape trajectories under the full Procrustes metric as

$$\hat{\mu}(t) = \operatorname{argmin}_{\omega \in \Omega} \frac{1}{n} \sum_{i=1}^n d_P^2(X_i(t), \omega),$$

and then analyzed the subject-specific Fréchet variance trajectories $d^2(X_i(t), \hat{\mu}(t)), t \in [0.5, 20], i = 1, 2, \dots, 232$ using the tools developed in this article. The sample Fréchet mean and the computation of the full Procrustes distance were carried out using the R package *shapes* (Dryden 2012). For details on existence and uniqueness of Fréchet means in shape spaces, see Le (1995, 1998).

Figure 8 shows the time evolution of population Fréchet variance, which captures the overall variability trends of the shape trajectories around the Fréchet mean trajectory. The periods of early childhood, that is from infancy to about 5 years of age, and the period starting at adolescence until adulthood, that is, between 13 and 20 years, exhibit greater variability around the Fréchet mean shapes. Figure 9 shows the first two dominant eigenfunctions, which explain 76.07% and 18.55% of the variability in the subject-specific Fréchet variance trajectories. The corresponding plot of the second FPC score versus the first FPC score shows that there is no systematic difference between boys and girls when comparing the shape trajectories under the full Procrustes metric.

To illustrate the growth shape patterns, we select four boys as indicated in the left panel of Figure 10 with the first FPC

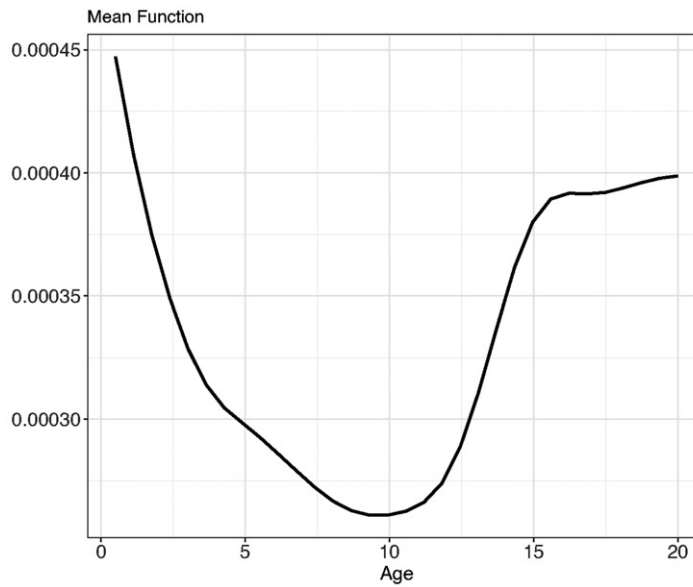


Figure 8. Sample mean function of the Fréchet variance trajectories of the time courses of shape data representing growth modalities.

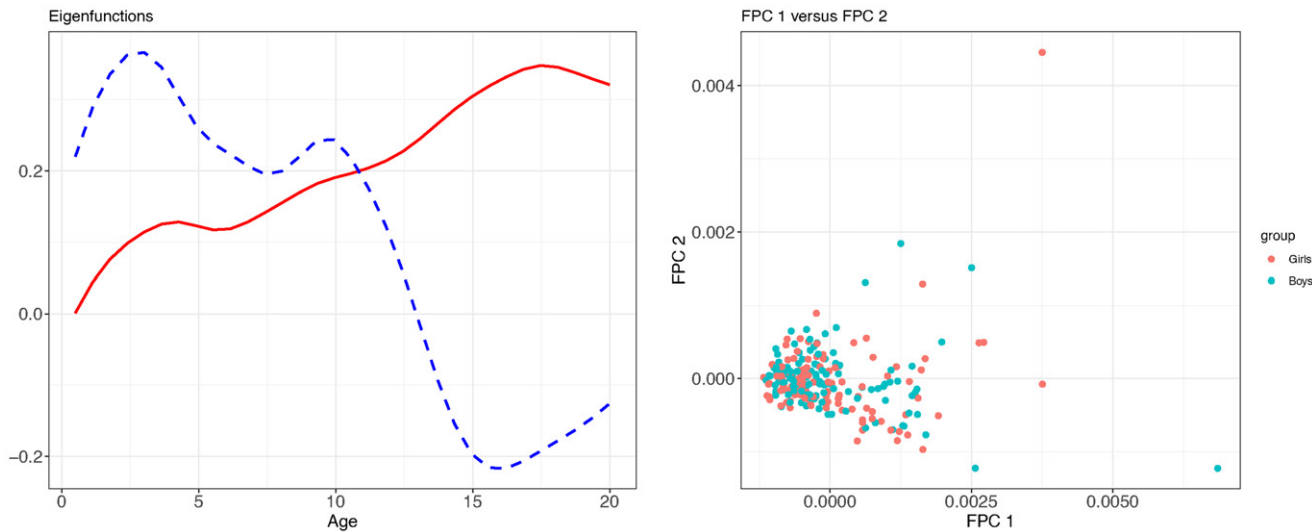


Figure 9. Left panel: First (solid red) and second (blue dashed) eigenfunction of the child-specific Fréchet variance trajectories, which explain 76.07% and 18.55% of variability, respectively. Right panel: Second versus first FPC scores of the child-specific Fréchet variance trajectories.

scores ranging from positive to zero to negative. A negative first FPC score of the subject-specific Fréchet variance trajectories loads negatively on the first eigenfunction illustrated in Figure 9, which highlights growth patterns that are closer to the sample Fréchet mean shape trajectory than average. A first FPC score closer to zero is associated with subjects that exhibit close to average variability around the sample Fréchet mean shape over time, while a positive first FPC score suggests greater than average variability around the mean Fréchet mean trajectory. The four boys “1,” “2,” “3,” and “4” illustrated in Figure 10 fit these prototypes. While “1” is closest to the sample Fréchet mean shape trajectory, “2” represents a child who shows typical deviation around the Fréchet mean shape over the years, whereas “3” exhibits larger than typical variation over the years and “4” stands out from the rest with an extremely high first FPC score. This pattern suggests that the first FPC captures differences in overall size of the children.

Figure 11 illustrates variability with respect to the second eigenfunction, which corresponds to a contrast between early and later years (Figure 9). For three selected boys highlighted in the left panel, all of whom have small first FPC scores, and whose second FPC scores vary from negative to zero to positive. For these boys, “5” tends to remain thinner during the teenage years as compared to “6” whose hip diameter gets wider during the teenage years, while “7” has the biggest growth in hip diameter among the three.

We also implemented empirical dynamics for these shape data. Figure 12 illustrates the slope function $\hat{\beta}(t)$ and the coefficient of determination $\hat{R}^2(t)$. The latter shows that dynamics in the distance trajectories between the mean shape trajectory and the subject-specific shape trajectories can be explained to a large extent by a first-order differential equation for the age range 5–15 years. The slope function $\hat{\beta}(t)$ is positive until 17 years of age and negative thereafter, indicating centrifugality

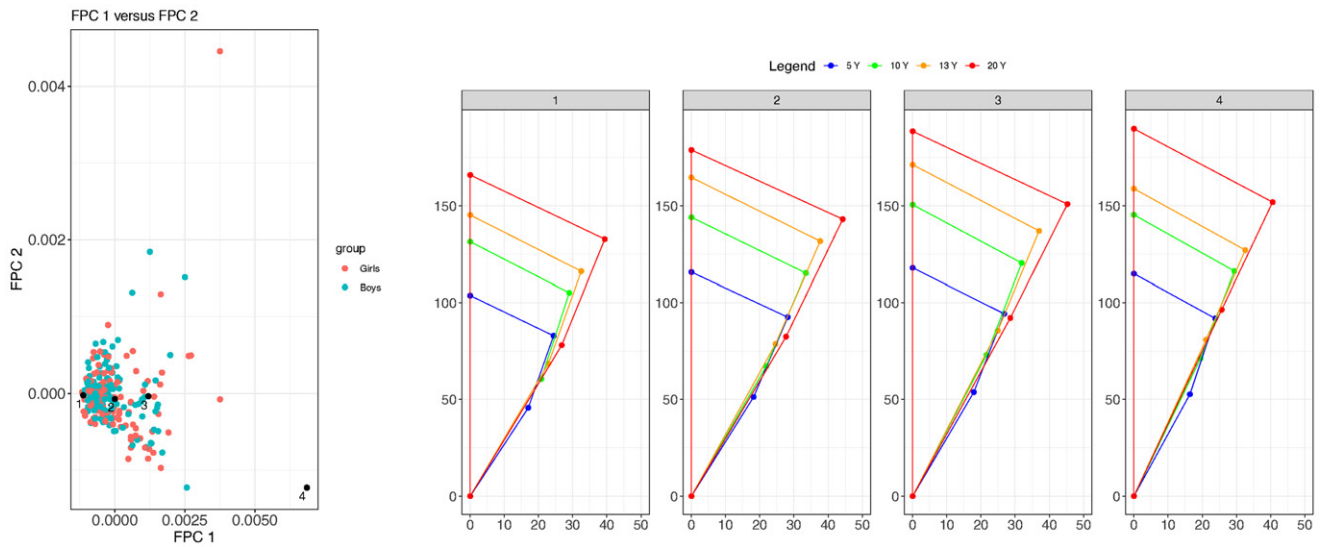


Figure 10. Left panel: Four selected children (black dots). Right panel: The shape trajectories $X_i(t)$, $t = 5, 10, 13$, and 20 for the four selected children in the left panel.

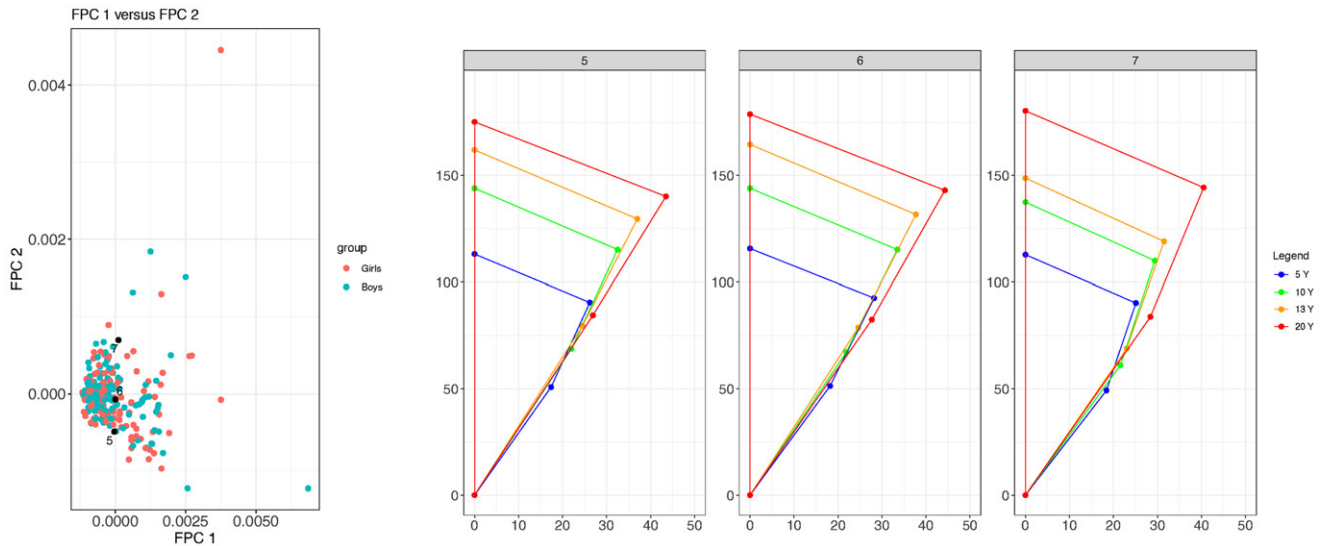


Figure 11. Left panel: Three selected children (black dots). Right panel: The shape trajectories $X_i(t)$, $t = 5, 10, 13$, and 20 for the three selected children in the left panel.

of the distance trajectories between infancy and late teenage years, where children's shapes diverge, and centripetality near adulthood.

5. Simulations

We illustrate our methods by simulations using samples of time-varying networks with 20 nodes, inspired by many real world networks that exhibit community structure, where communities are groups of nodes in a network that show increased within group connectivity and decreased between group connectivity. Existence of community structure is for example prevalent in traffic networks, particularly the bike networks that we study in our data applications, and also brain networks, social networks and many other areas where networks arise. We generated the time-varying networks as follows.

Step 1. Three groups of time-varying networks with 20 nodes differing in the community membership of the nodes were

generated. Indexing the nodes of the networks by $1, 2, \dots, 20$ and the communities by C_1, C_2, C_3, C_4 , and C_5 , the community membership composition of the nodes for the three groups of networks was as follows,

Group 1: Five communities, $C_1 = \{1, 2, 3, 4\}$, $C_2 = \{5, 6, 7, 8\}$, $C_3 = \{9, 10, 11, 12\}$, $C_4 = \{13, 14, 15, 16\}$ and $C_5 = \{17, 18, 19, 20\}$.

Group 2: Four communities, $C_1 = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $C_2 = \{9, 10, 11, 12\}$, $C_3 = \{13, 14, 15, 16\}$ and $C_4 = \{17, 18, 19, 20\}$.

Group 3: Three communities, $C_1 = \{1, 2, 3, 4\}$, $C_2 = \{5, 6, 7, 8\}$ and $C_3 = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$.

We let the community memberships of the nodes stay fixed in time, while the edge connectivity strengths $W_{jj'}(t)$ between the communities change with time. The time-varying connectivity weights $W_{jj'}(t)$ between communities C_j and $C_{j'}$, $j, j' \in \{1, 2, 3, 4, 5\}$, that we used when generating the random networks are illustrated in Figure 13. The intra-community connection strengths are higher than the inter-community strengths

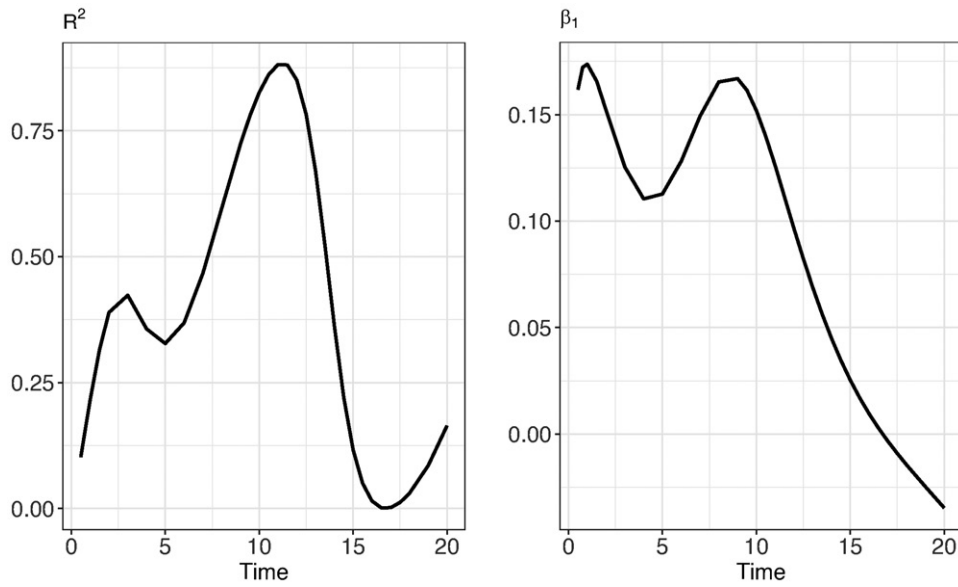


Figure 12. Smooth estimate of the coefficient of determination $R^2(t)$ (left) and the varying coefficient function $\beta(t)$ (right), capturing empirical dynamics of the distance trajectories of the longitudinal growth shape data.

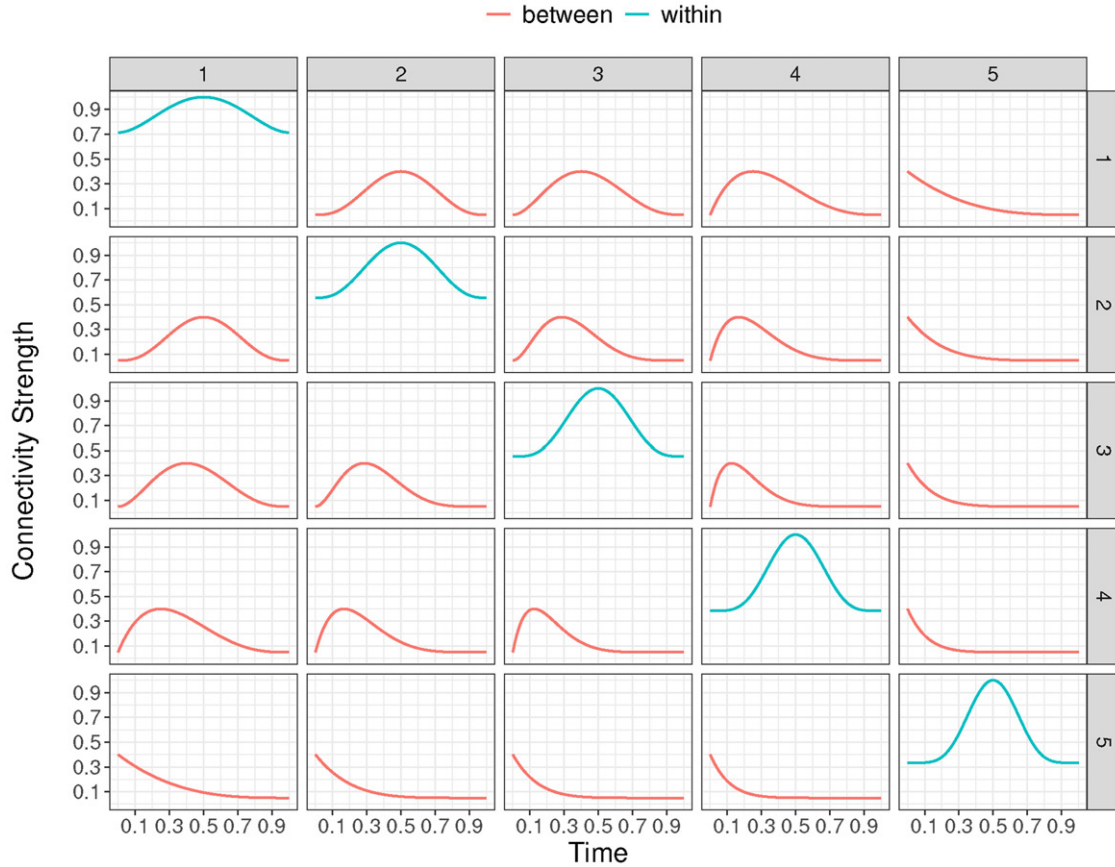


Figure 13. Time-varying connectivity weights between the five communities $C_1, C_2, \dots, C_5$.

over the entire time interval. Such dynamic connectivity patterns are encountered in brain networks (Calhoun et al. 2014), where densely connected brain regions form communities or hubs and inter-hub connectivity often exhibit changing patterns with age.

Step 2. The network adjacency matrices $A_i(t)$ are generated as follows:

$$(A_i(t))_{kl} = W_{jj'}(t) \left\{ \frac{1 + \sin(\pi(t + U_{i,kl})V_{i,kl})}{D} \right\}, \quad t \in [0, 1],$$

where C_j is the community membership of node k and $C_{j'}$ is the community membership of node l , $W_{jj'}(t)$ is the edge connectivity strength between nodes in communities C_j and $C_{j'}$, $U_{i,kl}$ follows $U(0, 1)$ and $V_{i,kl}$ is 1 if $j = j'$ and sampled

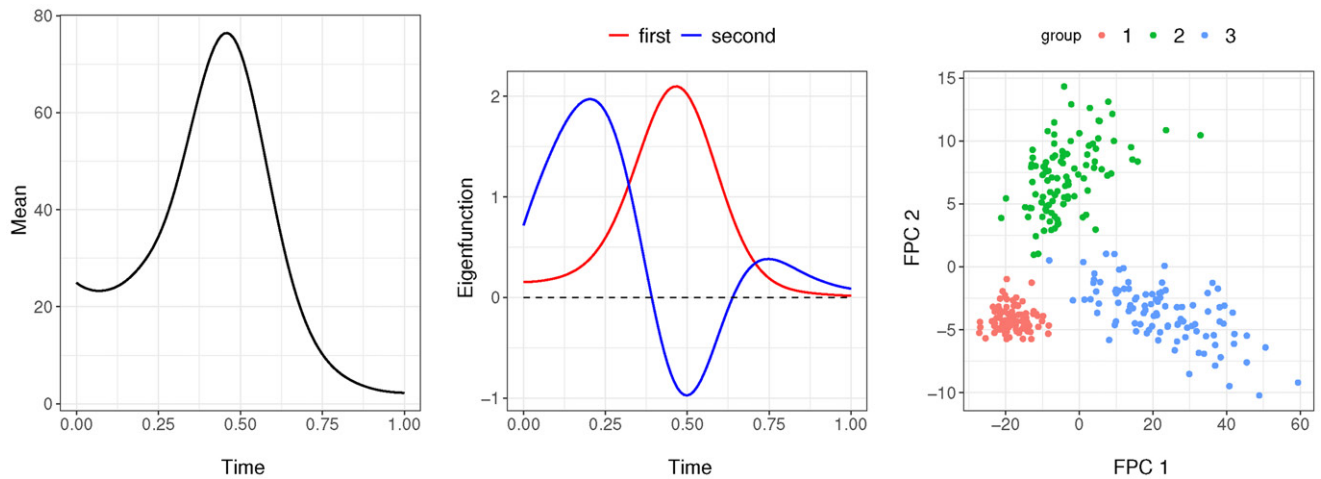


Figure 14. Mean function (left), eigenfunctions (middle) and the scatter plot of the second versus first FPC (right) obtained from FPCA of the distance trajectories for networks. The red line corresponds to the first eigenfunction and the blue line to the second.

uniformly from $\{5, 6, \dots, 15\}$ if $j \neq j'$. If $j = j'$, we set $D = 2$, otherwise $D = 4$. Here, $U_{i,kl}$ and $V_{i,kl}$ determine random phase and frequency shifts of the sine function which regulate at what times and how often the edge weights are zero. As $V_{i,kl}$ increases, so does the frequency of the times within $[0, 1]$ at which the edge weight is zero. The trajectories are represented as graph Laplacians

$$X_i(t) = D_i(t) - A_i(t),$$

where $D_i(t)$ is a diagonal matrix whose diagonal elements are equal to the sum of the corresponding row elements in $A_i(t)$. Adopting the Frobenius metric in the space of graph Laplacians, the Fréchet mean network at time t is the pointwise average of the graph Laplacians at time t , and we obtain the Frobenius distance trajectories of the individual subjects from the Fréchet mean trajectory.

Step 3. We carry out FPCA of the distance trajectories generated in Step 2.

The results are shown in Figure 14. The proposed method is seen to perform well in recovering the groups in the scatter plot of the second versus first functional principal component. Groups 1 and 2 are found to have closer cluster centers than Groups 1 and 3. This is explained by the fact that Group 2 is obtained from Group 1 by merging C_1 and C_2 in Group 1, which show more similarities than when merging C_3 , C_4 , and C_5 in Group 1 to form Group 3.

6. Discussion

We provide a framework for the analysis of time-varying object data, where the random objects can take values in a general metric space, by defining a generalized notion of mean function in the object space. The key to our approach is that FDA methodology can be applied to squared distance functions of the subject-specific curves from the mean function, elucidating the nature of the object time courses, including empirical dynamics, identifying clusters, and detecting extremes and potential outliers in a sample of object trajectories. Another important application is to determine time-specific ranks for subjects, in terms of the distance of the subject's trajectory from

the mean trajectory; such ranks have important applications in health monitoring, for example in neurodevelopment (Dosman, Andrews, and Goulden 2012). The FPCs that we obtain for time-varying random objects can also be used for regression analysis, where object time courses are responses or possibly predictors.

As pointed out by a referee, another possible interpretation of empirical dynamics can be gained with the notion of antimeans. For compact metric spaces, when the extreme values of the Fréchet function $E(d^2(Y, \omega))$ are attained in Ω , the set of maximizers of the Fréchet function forms a newly introduced notion of location parameter called the Fréchet *antimean* set (Patrangenaru, Guo, and Yao 2016a; Patrangenaru, Yao, and Guo 2016b; Wang, Patrangenaru, and Guo 2020). Just like Fréchet means, Fréchet antimeans can form useful statistics for describing and comparing samples of object data, as has been illustrated for samples of shapes of lily flowers (Patrangenaru, Yao, and Guo 2016b). For data that reside in compact manifolds where the notion of the data center is obscure, Fréchet antimeans could provide constructive data summaries. Hence, one could frame empirical dynamics with respect to not only the Fréchet means, but also the Fréchet antimeans, which might uncover insightful findings in terms of regression to the mean or the antimean, leading to multiple points of interest. In such situations, in the context of centrifugality, one might tend to move closer to the Fréchet antimean, for which *anticentripetality* might be a fitting term.

Our data examples include samples of time-varying univariate probability distributions and samples of time evolving networks. In practice the object functions may not be fully observed, but instead are observed on a more or less dense time grid, possibly with noise. In such situations, one can opt for smoothing the object trajectories if the observation grid is sufficiently dense. To implement preliminary smoothing and interpolation, Fréchet regression provides a possible option (Petersen and Müller 2019). While the role of smoothing individual trajectories in FDA is well understood in the Euclidean case (Hall and Van Keilegom 2007; Zhang and Chen 2007), it remains an open problem to investigate its properties in the much more general setting of longitudinal object data. An even bigger challenge that is also left for future research is the case

where measurement grids are sparse and irregular, a problem that was recently studied in Lin, Shao, and Yao (2020) and Dai, Lin, and Müller (2020) for data on Riemannian manifolds.

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Supplementary Material

The supplementary material contains a brief overview of the main theoretical developments in the paper, the proofs of Theorems 1 and 2 and auxiliary lemmas that were needed in the intermediate proof steps, additional illustrations for the Chicago Divvy bike data analysis and a discussion on examples of metric spaces that satisfy assumptions (A1)–(A5).

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